Stress and traction control in a Lagrange-reduced lattice finite-element model: nominal and Cauchy stress loading of periodic cells, and dead-load boundary tractions

Abstract

We describe how mechanical loads are prescribed in force (rather than displacement) in a two-dimensional lattice finite-element model in which the strain energy of every triangle is evaluated after Lagrange ($\mathrm{GL}(2,\mathbb{Z})$) reduction of its metric. Two settings are treated. For periodic cells, the macroscopic deformation gradient of the cell becomes an unknown conjugate to a prescribed macroscopic stress. A prescribed nominal (first Piola–Kirchhoff) stress is a dead load and leads to the minimization of an enthalpy in the joint space of internal coordinates and cell deformation. A prescribed true (Cauchy) stress is a follower load that admits no potential; it is enforced exactly by a fixed-point iteration on the nominal target, each pass being an enthalpy minimization. For finite specimens, dead-load nominal tractions on boundary segments and rigid load-controlled grips are introduced through external work potentials. For both settings we give the equilibrium equations, the work-conjugate strain measures used to detect plastic strain bursts, the treatment of rigid-body modes and of the gauge rotation, and the detection of the limit load past which no equilibrium exists. Finally, Newton-type alternatives based on the Cauchy–Born, relaxed (Schur-complement) and full Hessian tangents are formulated and compared with the first-order method.

1 Discrete energy

The reference configuration is a triangulation of the lattice sites $\{\mathbf{X}_a\}$. Each triangle e with vertices $a \in e$ carries constant shape-function gradients ∇N_a^e, so that its deformation gradient is

$$\mathbf{F}_e = \sum_{a\in e} \mathbf{x}_a \otimes \nabla N_a^e, \qquad \mathbf{C}_e = \mathbf{F}_e^{\mathsf{T}}\mathbf{F}_e. \tag{1}$$

The strain-energy density per unit reference area is obtained from a lattice (pair or many-body) energy Φ evaluated on the reduced metric,

$$W(\mathbf{C}) = \frac{1}{A_{\mathrm{cell}}}\,\Phi\Big(\mathbf{m}^{\mathsf{T}}\mathbf{C}\,\mathbf{m}\Big), \qquad \mathbf{m} = \mathbf{m}(\mathbf{C}) \in \mathrm{GL}(2,\mathbb{Z}), \tag{2}$$

where $\mathbf{m}(\mathbf{C})$ is the Lagrange reduction of $\mathbf{C}$ and A_{cell} is the area of the lattice unit cell, which makes W an energy per unit area. By construction W is invariant under $\mathbf{C} \mapsto \mathbf{n}^{\mathsf{T}}\mathbf{C}\mathbf{n}$ for all $\mathbf{n} \in \mathrm{GL}(2,\mathbb{Z})$, i.e. under lattice-invariant shears, which is what allows dislocations to nucleate and glide without any reconnection of the underlying lattice. The first Piola–Kirchhoff stress of a triangle is

$$\mathbf{P}_e = \frac{\partial W}{\partial \mathbf{F}}(\mathbf{F}_e) = \frac{2}{A_{\mathrm{cell}}}\,\mathbf{F}_e\,\mathbf{m}\,\frac{\partial \Phi}{\partial \mathbf{C}}\big(\mathbf{m}^{\mathsf{T}}\mathbf{C}_e\mathbf{m}\big)\,\mathbf{m}^{\mathsf{T}}, \tag{3}$$

and the total elastic energy is

$$E_{\mathrm{el}} = \sum_e A_e\, W(\mathbf{C}_e), \qquad \frac{\partial E_{\mathrm{el}}}{\partial \mathbf{x}_a} = \sum_{e\ni a} A_e\,\mathbf{P}_e\,\nabla N_a^e. \tag{4}$$

Here A_e is the reference area of triangle e. Because ∇N_a^e is fixed by the reference element, A_e must be the area of that reference element, $A_e = \frac{1}{2}\,|\det[\nabla N_2^e, \nabla N_3^e]|^{-1}$, and not the area of the triangle formed by the reference positions of the current vertices: after a mesh reconnection the two differ, and using the latter corrupts the energy and the stress of the reconnected triangles.

2 Periodic cells: kinematics and macroscopic stress

2.1 Cell deformation and internal coordinates

A periodic cell has reference area A_0 and is deformed by a homogeneous macroscopic deformation gradient $\bar{\mathbf{F}}$. Atomic positions are written as

$$\mathbf{x}_a = \bar{\mathbf{F}}\,\mathbf{w}_a, \tag{5}$$

where $\mathbf{w}_a$ are internal (cell) coordinates. A triangle that straddles the cell boundary uses periodic images of its vertices, $\mathbf{x}_a + \bar{\mathbf{F}}\,\mathbf{T}_a^e$, with $\mathbf{T}_a^e$ a fixed translation by a reference period vector. Substituting in (1),

$$\mathbf{F}_e = \bar{\mathbf{F}}\,\mathbf{G}_e, \qquad \mathbf{G}_e = \sum_{a\in e} \left(\mathbf{w}_a + \mathbf{T}_a^e\right) \otimes \nabla N_a^e, \tag{6}$$

where $\mathbf{G}_e$ does not depend on $\bar{\mathbf{F}}$. The energy is therefore a function $E_{\mathrm{el}}(\mathbf{w}, \bar{\mathbf{F}})$, and its partial derivatives at fixed $\mathbf{w}$ and at fixed $\bar{\mathbf{F}}$ are

$$\frac{\partial E_{\mathrm{el}}}{\partial \mathbf{w}_a} = \bar{\mathbf{F}}^{\mathsf{T}}\,\frac{\partial E_{\mathrm{el}}}{\partial \mathbf{x}_a}, \tag{7}$$

$$\frac{\partial E_{\mathrm{el}}}{\partial \bar{\mathbf{F}}} = \sum_e A_e\,\mathbf{P}_e\,\mathbf{G}_e^{\mathsf{T}} = \Big(\sum_e A_e\,\mathbf{P}_e\,\mathbf{F}_e^{\mathsf{T}}\Big)\bar{\mathbf{F}}^{-\mathsf{T}}. \tag{8}$$

The matrix $\mathbf{V} = \sum_e A_e \mathbf{P}_e \mathbf{F}_e^{\mathsf{T}}$ is the (Kirchhoff) virial of the cell. It is accumulated during the same element loop that computes the energy and the nodal forces, so the derivative with respect to the cell costs nothing extra.

2.2 Macroscopic stresses

The macroscopic nominal and Cauchy stresses are defined by work conjugacy with the cell deformation,

$$\bar{\mathbf{P}} = \frac{1}{A_0}\,\frac{\partial E_{\mathrm{el}}}{\partial \bar{\mathbf{F}}} = \frac{1}{A_0}\,\mathbf{V}\,\bar{\mathbf{F}}^{-\mathsf{T}}, \qquad \bar{\boldsymbol{\sigma}} = \frac{1}{\bar{J}}\,\bar{\mathbf{P}}\,\bar{\mathbf{F}}^{\mathsf{T}} = \frac{\mathbf{V}}{A_0\,\bar{J}}, \qquad \bar{J} = \det\bar{\mathbf{F}}. \tag{9}$$

Since $\mathbf{P}_e\mathbf{F}_e^{\mathsf{T}} = J_e\boldsymbol{\sigma}_e$ and $A_eJ_e = a_e$ is the current area of the triangle, $\bar{\boldsymbol{\sigma}} = \sum_e a_e\boldsymbol{\sigma}_e/(A_0\bar{J})$ is the current-area average of the element Cauchy stresses, provided $A_0\bar{J}$ is the current area of the cell. At atomic equilibrium, $\partial E_{\mathrm{el}}/\partial\mathbf{w} = 0$, the value of $\bar{\mathbf{P}}$ does not depend on which variables are held fixed while $\bar{\mathbf{F}}$ is varied (Hill–Mandel condition).

Cell reference area. The work done on the cell per unit change of $\bar{\mathbf{F}}$ is $A_0\,\bar{\mathbf{P}} : \mathrm{d}\bar{\mathbf{F}}$, with A_0 the reference area of the *periodic cell*. The triangles tile the current cell, so we compute

$$A_0 = \frac{1}{\det\bar{\mathbf{F}}}\sum_e a_e. \tag{10}$$

This is in general different from $\sum_e A_e$. For instance, a dislocation dipole inserted in a periodic cell adds material half-planes between the two cores, so the summed material reference area exceeds the cell area (in one of our test cells, by 586 against 576). Using $\sum_e A_e$ in place of (10) produces a spurious offset between the prescribed and the measured stress.

2.3 Rotation gauge

The energy is objective, $E_{\mathrm{el}}(\mathbf{w}, \mathbf{R}\bar{\mathbf{F}}) = E_{\mathrm{el}}(\mathbf{w}, \bar{\mathbf{F}})$ for every rotation $\mathbf{R}$. Every $\bar{\mathbf{F}}$ with positive determinant has a unique decomposition $\bar{\mathbf{F}} = \mathbf{R}\,\bar{\mathbf{F}}_{\triangle}$ with $\bar{\mathbf{F}}_{\triangle}$ upper triangular with positive diagonal (the QR decomposition), so we fix the gauge

$$\bar{\mathbf{F}} = \begin{pmatrix} \bar{F}_{11} & \bar{F}_{12} \\ 0 & \bar{F}_{22} \end{pmatrix}, \qquad \bar{F}_{11}, \bar{F}_{22} > 0. \tag{11}$$

The cell then has three kinematic degrees of freedom $(\bar{F}_{11}, \bar{F}_{12}, \bar{F}_{22})$, conjugate to $(\bar{P}_{11}, \bar{P}_{12}, \bar{P}_{22})$. The fourth component $\bar{P}_{21}$ is not an independent unknown: objectivity implies the rotational balance $\bar{\mathbf{P}}\,\bar{\mathbf{F}}^{\mathsf{T}} = (\bar{\mathbf{P}}\,\bar{\mathbf{F}}^{\mathsf{T}})^{\mathsf{T}}$, which in the gauge (11) reads

$$\bar{P}_{21}\,\bar{F}_{11} = \bar{P}_{11}\,\bar{F}_{12} + \bar{P}_{12}\,\bar{F}_{22}. \tag{12}$$

Proposition 1. *Let $\bar{\mathbf{F}}$ satisfy* (11) *and let $\bar{\mathbf{P}}$ and $\mathbf{P}^{\mathrm{t}}$ both satisfy the rotational balance, $\bar{\mathbf{P}}\bar{\mathbf{F}}^{\mathsf{T}}$ and $\mathbf{P}^{\mathrm{t}}\bar{\mathbf{F}}^{\mathsf{T}}$ symmetric. If $\bar{P}_{ij} = P^{\mathrm{t}}_{ij}$ for $(ij) \in \{11, 12, 22\}$, then $\bar{\mathbf{P}} = \mathbf{P}^{\mathrm{t}}$.*

Proof. $\mathbf{D} = \bar{\mathbf{P}} - \mathbf{P}^{\mathrm{t}}$ has the single non-zero entry $D_{21} = d$, and $\mathbf{D}\bar{\mathbf{F}}^{\mathsf{T}}$ has the single non-zero entry $(\mathbf{D}\bar{\mathbf{F}}^{\mathsf{T}})_{21} = d\,\bar{F}_{11}$. It is symmetric only if $d\bar{F}_{11} = 0$, i.e. $d = 0$ since $\bar{F}_{11} > 0$. □

In particular, if the target is a Cauchy stress, $\mathbf{P}^{\mathrm{t}} = \bar{J}\boldsymbol{\sigma}^{\mathrm{t}}\bar{\mathbf{F}}^{-\mathsf{T}}$, then $\mathbf{P}^{\mathrm{t}}\bar{\mathbf{F}}^{\mathsf{T}} = \bar{J}\boldsymbol{\sigma}^{\mathrm{t}}$ is symmetric, and matching the three free nominal components gives the *complete* Cauchy stress $\bar{\boldsymbol{\sigma}} = \boldsymbol{\sigma}^{\mathrm{t}}$, including the component conjugate to the gauged-out rotation.

3 Nominal (Piola) stress control of a periodic cell

3.1 Enthalpy

Let $\mathcal{S} \subseteq \{11, 12, 22\}$ be the set of stress-driven cell components; the others are strain-driven (prescribed). For a prescribed nominal stress $\mathbf{P}^{\mathrm{t}}$ the external work is linear in $\bar{\mathbf{F}}$, the load is dead, and the equilibria are the stationary points of the enthalpy

$$H(\mathbf{w}, \bar{\mathbf{F}}) = E_{\mathrm{el}}(\mathbf{w}, \bar{\mathbf{F}}) - A_0 \sum_{(ij)\in\mathcal{S}} P^{\mathrm{t}}_{ij}\big(\bar{F}_{ij} - \delta_{ij}\big). \tag{13}$$

With (7)–(9), the stationarity conditions are

$$\frac{\partial H}{\partial \mathbf{w}_a} = \bar{\mathbf{F}}^{\mathsf{T}}\,\frac{\partial E_{\mathrm{el}}}{\partial \mathbf{x}_a} = \mathbf{0} \qquad \text{(atomic equilibrium)}, \tag{14}$$

$$\frac{\partial H}{\partial \bar{F}_{ij}} = A_0\big(\bar{P}_{ij} - P^{\mathrm{t}}_{ij}\big) = 0, \quad (ij) \in \mathcal{S} \qquad \text{(macroscopic equilibrium)}. \tag{15}$$

Since the work term is linear, $\nabla^2 H = \nabla^2 E_{\mathrm{el}}$ in the joint variables $(\mathbf{w}, \bar{\mathbf{F}}_{\mathcal{S}})$: an equilibrium is stable when the elastic Hessian, with the cell components in $\mathcal{S}$ free, is positive definite. This is a weaker condition than stability at fixed $\bar{\mathbf{F}}$ (strain control), and the difference is the source of the strain bursts discussed in Section 6.

3.2 Loading presets

Table 1 lists the loadings implemented. A "shear" loading prescribes $P^{\mathrm{t}}_{12} = \alpha$ with $P^{\mathrm{t}}_{11} = P^{\mathrm{t}}_{22} = 0$, all three components being free; uniaxial loadings prescribe one normal component. A mixed loading keeps $\bar{F}_{12} = \alpha$ prescribed and relaxes the two normal components to zero nominal stress. An optional zero-stress relaxation of all three components at $\alpha = 0$ removes the residual stress of the initial microstructure (e.g. of an inserted dipole) before loading.

Loading	free $\mathcal{S}$	target	conjugate strain ε
nominal shear	11, 12, 22	$P_{12}=\alpha$, $P_{11}=P_{22}=0$	$\bar{F}_{12}-\bar{F}_{12}^{(0)}$
nominal uniaxial y	11, 12, 22	$P_{22}=\alpha$, others 0	$\bar{F}_{22}-\bar{F}_{22}^{(0)}$
nominal uniaxial x	11, 12, 22	$P_{11}=\alpha$, others 0	$\bar{F}_{11}-\bar{F}_{11}^{(0)}$
Cauchy shear	11, 12, 22	$\sigma_{12}=\sigma_{21}=\alpha$, $\sigma_{11}=\sigma_{22}=0$	$\bar{F}_{12}-\bar{F}_{12}^{(0)}$
Cauchy uniaxial y	11, 12, 22	$\sigma_{22}=\alpha$, others 0	$\bar{F}_{22}-\bar{F}_{22}^{(0)}$
Cauchy uniaxial x	11, 12, 22	$\sigma_{11}=\alpha$, others 0	$\bar{F}_{11}-\bar{F}_{11}^{(0)}$
mixed	11, 22	$\bar{F}_{12}=\alpha$, $P_{11}=P_{22}=0$	—
zero-stress relaxation	11, 12, 22	$\mathbf{P}=\mathbf{0}$ at $\alpha=0$	—

Table 1: Loadings of the periodic cell. $\bar{\mathbf{F}}^{(0)}$ is the cell after the initial relaxation step, the origin of the strain axis.

3.3 Nominal shear is not a pure Cauchy shear

The nominal and Cauchy shear loadings differ by more than a change of measure. Take $\bar{\mathbf{P}} = \tau\,\mathbf{e}_1\otimes\mathbf{e}_2 + \bar{P}_{21}\,\mathbf{e}_2\otimes\mathbf{e}_1$ (i.e. $\bar{P}_{11}=\bar{P}_{22}=0$, $\bar{P}_{12}=\tau$), with $\bar{P}_{21}=\tau\bar{F}_{22}/\bar{F}_{11}$ from (12). With $\bar{J}=\bar{F}_{11}\bar{F}_{22}$, (9) gives

$$\bar{\boldsymbol{\sigma}} = \frac{\tau}{\bar{F}_{11}\bar{F}_{22}}\begin{pmatrix}\bar{F}_{12} & \bar{F}_{22}\\ \bar{F}_{22} & 0\end{pmatrix}, \qquad \bar{\sigma}_{12}=\frac{\tau}{\bar{F}_{11}}, \quad \bar{\sigma}_{11}=\frac{\tau\,\bar{F}_{12}}{\bar{F}_{11}\bar{F}_{22}}. \tag{16}$$

Under nominal shear the cell therefore carries a normal Cauchy stress $\bar{\sigma}_{11}\approx\tau\gamma$ that grows with the shear $\gamma=\bar{F}_{12}$. At the shear strains reached before dislocation nucleation ($\gamma\sim 0.1$–1) this is not small, and the resolved shear stress differs from the one obtained under a pure shear stress.

4 Cauchy (true) stress control of a periodic cell

4.1 The Cauchy target is a follower load

A prescribed Cauchy stress $\boldsymbol{\sigma}^{\mathrm{t}}$ corresponds to the deformation-dependent nominal target

$$\mathbf{P}^{\mathrm{t}}(\bar{\mathbf{F}}) = \bar{J}\,\boldsymbol{\sigma}^{\mathrm{t}}\,\bar{\mathbf{F}}^{-\mathsf{T}}. \tag{17}$$

In the gauge (11), with $\bar{\mathbf{F}}^{-\mathsf{T}}=\begin{pmatrix}1/\bar{F}_{11} & 0\\ -\bar{F}_{12}/(\bar{F}_{11}\bar{F}_{22}) & 1/\bar{F}_{22}\end{pmatrix}$, the three loadings of Table 1 give the explicit targets

$$\boldsymbol{\sigma}^{\mathrm{t}}=\tau(\mathbf{e}_1\otimes\mathbf{e}_2+\mathbf{e}_2\otimes\mathbf{e}_1): \quad P^{\mathrm{t}}_{11}=-\tau\bar{F}_{12}, \quad P^{\mathrm{t}}_{12}=\tau\bar{F}_{11}, \quad P^{\mathrm{t}}_{22}=0, \tag{18}$$

$$\boldsymbol{\sigma}^{\mathrm{t}}=s\,\mathbf{e}_2\otimes\mathbf{e}_2: \quad P^{\mathrm{t}}_{11}=0, \quad P^{\mathrm{t}}_{12}=0, \quad P^{\mathrm{t}}_{22}=s\,\bar{F}_{11}, \tag{19}$$

$$\boldsymbol{\sigma}^{\mathrm{t}}=s\,\mathbf{e}_1\otimes\mathbf{e}_1: \quad P^{\mathrm{t}}_{11}=s\,\bar{F}_{22}, \quad P^{\mathrm{t}}_{12}=0, \quad P^{\mathrm{t}}_{22}=0. \tag{20}$$

A pure Cauchy shear thus requires a compressive nominal normal stress $-\tau\bar{F}_{12}$ on the cell, which cancels the normal Cauchy stress of (16). The work one-form $\omega=A_0\sum_{\mathcal{S}}P^{\mathrm{t}}_{ij}(\bar{\mathbf{F}})\,\mathrm{d}\bar{F}_{ij}$ is not closed: for (18), $\omega=A_0\tau(-\bar{F}_{12}\,\mathrm{d}\bar{F}_{11}+\bar{F}_{11}\,\mathrm{d}\bar{F}_{12})$ and $\mathrm{d}\omega=2A_0\tau\,\mathrm{d}\bar{F}_{11}\wedge\mathrm{d}\bar{F}_{12}\neq 0$; likewise $\mathrm{d}\omega=A_0 s\,\mathrm{d}\bar{F}_{11}\wedge\mathrm{d}\bar{F}_{22}$ for (19) and $-A_0 s\,\mathrm{d}\bar{F}_{11}\wedge\mathrm{d}\bar{F}_{22}$ for (20). A prescribed deviatoric Cauchy stress therefore has no work potential, and the equilibria cannot be obtained by minimizing a single enthalpy. (A hydrostatic target $\boldsymbol{\sigma}^{\mathrm{t}}=-p\mathbf{I}$ is the exception: $\omega=-pA_0\,\mathrm{d}\bar{J}$ is exact.)

4.2 Fixed-point iteration on the nominal target

The equilibrium conditions under Cauchy control are (14) together with

$$\bar{P}_{ij}(\mathbf{w},\bar{\mathbf{F}})=P^{\mathrm{t}}_{ij}(\bar{\mathbf{F}}), \qquad (ij)\in\mathcal{S}. \tag{21}$$

Algorithm 1 Cauchy stress control of a periodic cell at load level α

Require: configuration $(\mathbf{w}, \bar{\mathbf{F}})$ from the predictor; Cauchy target $\boldsymbol{\sigma}^{\mathrm{t}}(\alpha)$

tol $\leftarrow 10^{-11} + 10^{-9} \max_{ij} |\sigma^{\mathrm{t}}_{ij}|$
for $k = 1, \dots, 60$ **do**
  $\mathbf{P}^{\mathrm{t}}_k \leftarrow \det \bar{\mathbf{F}}\ \boldsymbol{\sigma}^{\mathrm{t}}\, \bar{\mathbf{F}}^{-\mathsf{T}}$
  $(\mathbf{w}, \bar{\mathbf{F}}) \leftarrow \arg\min H(\,\cdot\,; \mathbf{P}^{\mathrm{t}}_k)$ ▷ L-BFGS from the current state, Eq. (13)
  $\mathbf{P}^{\mathrm{t}}_{k+1} \leftarrow \det \bar{\mathbf{F}}\ \boldsymbol{\sigma}^{\mathrm{t}}\, \bar{\mathbf{F}}^{-\mathsf{T}}$
  if $\max_{ij} |P^{\mathrm{t}}_{k+1,ij} - P^{\mathrm{t}}_{k,ij}| \leq$ tol **then**
    return converged
  end if
  if a stress-driven $\bar{F}_{ij}$ moved by more than $\Delta F_{\max}$ in this load step **then**
    return limit load ▷ Section 6.4
  end if
end for

For any frozen value $\mathbf{P}^{\mathrm{t}}_k$ of the target, (14)–(15) are the stationarity conditions of the dead-load enthalpy (13). We therefore solve (21) by the fixed-point iteration of Algorithm 1: freeze the nominal target at the current cell, minimize the enthalpy, re-evaluate the target at the relaxed cell, and repeat until the target no longer changes. Each pass is warm-started from the previous minimizer, so after the first pass the remaining passes are short.

Convergence. Linearize about the solution $\bar{\mathbf{F}}^*$. Let $\mathbb{K}$ be the relaxed macroscopic tangent of the cell, i.e. $\mathrm{d}\bar{\mathbf{P}} = \mathbb{K} : \mathrm{d}\bar{\mathbf{F}}$ with the atoms at equilibrium, restricted to the components in $\mathcal{S}$. A perturbation $\delta\bar{\mathbf{F}}_k$ of the cell changes the frozen target by $\delta\mathbf{P}^{\mathrm{t}}_k = \mathbb{L} : \delta\bar{\mathbf{F}}_k$ with $\mathbb{L} = \partial\mathbf{P}^{\mathrm{t}}/\partial\bar{\mathbf{F}}$, and the next minimization returns $\delta\bar{\mathbf{F}}_{k+1} = \mathbb{K}^{-1}\mathbb{L}\,\delta\bar{\mathbf{F}}_k$. From (18)–(20), $\|\mathbb{L}\| = O(|\boldsymbol{\sigma}^{\mathrm{t}}|)$, so the iteration contracts with ratio

$$\rho = \|\mathbb{K}^{-1}\mathbb{L}\| \sim \frac{|\boldsymbol{\sigma}^{\mathrm{t}}|}{\mu_{\mathrm{eff}}}, \tag{22}$$

where μ_{eff} is the smallest relaxed modulus of the cell. For stresses well below the modulus the target typically settles to the tolerance in about six passes. As the load approaches the limit load, $\mathbb{K}$ becomes singular, ρ grows without bound and the iteration stops converging; this is detected by the limit-load guard rather than by the pass limit.

Exactness and the measured stress. At convergence, (15) holds with $\mathbf{P}^{\mathrm{t}} = \bar{J}\boldsymbol{\sigma}^{\mathrm{t}}\bar{\mathbf{F}}^{-\mathsf{T}}$ evaluated at the final cell, and Proposition 1 gives $\bar{\boldsymbol{\sigma}} = \boldsymbol{\sigma}^{\mathrm{t}}$ for all four components. In practice the cell Cauchy stress computed from the virial, $\bar{\boldsymbol{\sigma}} = \mathbf{V}/(A_0\bar{J})$, matches the target to $\sim 10^{-8}$ in the controlled component, and the nominally zero components are of the same order.

Comparing states under Cauchy control. The reconnection cascade (Section 6) accepts a new topology when it lowers the enthalpy at fixed load. Under Cauchy control the two states are in equilibrium with different nominal targets, $\mathbf{P}^{\mathrm{t}}_0 = \mathbf{P}^{\mathrm{t}}(\bar{\mathbf{F}}_0)$ and $\mathbf{P}^{\mathrm{t}}_1 = \mathbf{P}^{\mathrm{t}}(\bar{\mathbf{F}}_1)$, and there is no potential whose difference would measure the work of the load. Comparing $H(\bar{\mathbf{F}}_1; \mathbf{P}^{\mathrm{t}}_1)$ with $H(\bar{\mathbf{F}}_0; \mathbf{P}^{\mathrm{t}}_0)$ would include the spurious term $-A_0(\mathbf{P}^{\mathrm{t}}_1 - \mathbf{P}^{\mathrm{t}}_0) : (\bar{\mathbf{F}} - \mathbf{I})$. We instead evaluate the work of the load with the trapezoid rule, $A_0\,\frac{1}{2}(\mathbf{P}^{\mathrm{t}}_0 + \mathbf{P}^{\mathrm{t}}_1) : (\bar{\mathbf{F}}_1 - \bar{\mathbf{F}}_0)$, so that

$$\Delta\Pi = E_{\mathrm{el}}(1) - E_{\mathrm{el}}(0) - A_0\,\tfrac{1}{2}(\mathbf{P}^{\mathrm{t}}_0 + \mathbf{P}^{\mathrm{t}}_1) : (\bar{\mathbf{F}}_1 - \bar{\mathbf{F}}_0) = \Delta H + A_0\,(\mathbf{P}^{\mathrm{t}}_1 - \mathbf{P}^{\mathrm{t}}_0) : \big(\tfrac{1}{2}(\bar{\mathbf{F}}_0 + \bar{\mathbf{F}}_1) - \mathbf{I}\big), \tag{23}$$

summed over the free components; it reduces to ΔH under nominal control. A rejected reconnection restores both the cell and the target.

Remark 1. *Each pass minimizes the frozen-load enthalpy, so its stability test uses $\nabla^2 E_{\mathrm{el}}$ and omits the load stiffness $-A_0\,\partial\mathbf{P}^{\mathrm{t}}/\partial\bar{\mathbf{F}}$ of the follower load. That term is of relative order $|\boldsymbol{\sigma}^{\mathrm{t}}|/\mu_{\mathrm{eff}}$, the same small parameter as in* (22).

5 Numerical treatment of the cell variables

The unknowns of a stress-controlled relaxation are $\mathbf{z} = (\mathbf{w},\ \sqrt{A_0}\,\bar{F}_{ij})$ for $(ij) \in \mathcal{S}$. Scaling the cell components by $\sqrt{A_0}$ gives every unknown the dimension of a length, so that the gradient components

$$\frac{\partial H}{\partial(\sqrt{A_0}\,\bar{F}_{ij})} = \frac{1}{\sqrt{A_0}}\Big(\big[\mathbf{V}\bar{\mathbf{F}}^{-\mathsf{T}}\big]_{ij} - A_0 P^{\mathrm{t}}_{ij}\Big) = \sqrt{A_0}\,\big(\bar{P}_{ij} - P^{\mathrm{t}}_{ij}\big) \tag{24}$$

have the same scaling as the atomic forces, and the limited-memory quasi-Newton (L-BFGS) Hessian estimate is well conditioned across the two blocks. Positions are recovered as $\mathbf{x}_a = \bar{\mathbf{F}}\mathbf{w}_a$, and the periodic image translations follow $\bar{\mathbf{F}}$ in every function evaluation.

Lagrange reduction makes the homogeneous energy bounded and periodic along lattice-invariant shears of the cell, while the work term in (13) grows linearly in $\bar{F}_{12}$. For any $\tau \neq 0$, H is therefore unbounded below: all equilibria under stress control are local minima, separated from runaway shearing of the cell by the barrier of the energy landscape. To keep the line search from extrapolating across that barrier, the step length of the minimizer is capped (to 0.5 in units of the lattice parameter in $\mathbf{z}$).

When the stiffness preconditioner is selected, the same minimization is carried out in preconditioned variables, with the atom–cell stiffness factorized as a bordered sparse Cholesky factor. The construction is given in Section 8.4.

6 Load stepping, avalanches and the limit load

6.1 Schedule and predictor

The load parameter α runs through a schedule $\alpha_0, \alpha_1, \ldots$; under stress control α is the prescribed stress component (P_{12}, P_{22}, P_{11}, σ_{12}, σ_{22} or σ_{11}). At each step a predictor is applied before relaxation. Strain-driven cell components follow the schedule from the relaxed initial cell $\bar{\mathbf{F}}^{(0)}$. Stress-driven components are extrapolated from the last two converged cells,

$$\bar{F}^{\mathrm{pred}}_{ij} = \bar{F}^{(n)}_{ij} + \frac{\Delta\alpha_{n+1}}{\Delta\alpha_n}\big(\bar{F}^{(n)}_{ij} - \bar{F}^{(n-1)}_{ij}\big), \tag{25}$$

but only when step n was smooth (no reconnection and no strain burst); otherwise the predictor is the last converged cell. The relaxation then follows (Sections 3–4), followed by the cascade of Delaunay reconnections of the mesh, each accepted only if it lowers the energy functional being minimized, i.e. H under stress control.

6.2 Initial relaxation at a prescribed shear strain

A perfect periodic crystal under stress control loads elastically until the limit load, where the cell shears away without an intermediate defected state. To start the stress-controlled run from a dislocated microstructure, the first step can be performed under *strain* control at a non-zero shear $\bar{F}_{12} = \gamma_0$, with positional noise, together with the reconnection cascade. Beyond the homogeneous instability of the lattice (for the potentials studied here, $\gamma_0 \gtrsim 0.14$) and in a sufficiently large cell, this relaxation nucleates a population of dislocations. The relaxed shear stress σ_0 of that state is then measured, and the stress schedule is $\alpha_k = \sigma_0 + k\,\Delta\alpha$, $k \geq 1$, so that the first stress step continues from the stress the defected crystal already carries, with no jump in load.

6.3 Strain bursts

Under strain control an avalanche appears as a drop of stress at fixed strain. Under force control the stress is fixed and an avalanche appears instead as a burst of the work-conjugate strain ε

(Table 1; for boundary tractions, Eq. (35) below). Let $\Delta\varepsilon_{\mathrm{el}}/\Delta\alpha_{\mathrm{el}}$ be the secant compliance of the last step that showed no burst. Step n is flagged as an avalanche when

$$\frac{|\Delta\varepsilon_n| - |\Delta\varepsilon_n^{\mathrm{el}}|}{|\Delta\varepsilon_n^{\mathrm{el}}|} \;\geq\; \theta, \qquad \Delta\varepsilon_n^{\mathrm{el}} = \Delta\varepsilon_{\mathrm{el}}\,\frac{\Delta\alpha_n}{\Delta\alpha_{\mathrm{el}}}, \tag{26}$$

and, when remeshing is enabled, the reconnection cascade of the step was accepted and lowered the enthalpy below its value after the continuous relaxation. The left-hand side of (26) is the fractional strain burst, the force-control analogue of the fractional stress drop.

Energy reported. Under force control the functional minimized is $H = E_{\mathrm{el}} - W_{\mathrm{ext}}$, which decreases monotonically along a loading path dominated by the work term. The elastic energy E_{el} is reported separately: it drops when dislocations annihilate or leave the cell and builds up as stress is stored.

6.4 Limit load

Past the limit load (the yield or Peierls stress of the current microstructure) the dead load has no equilibrium, and the relaxation produces a runaway configuration. The step is declared a runaway when, within one load step, a stress-driven cell component changes by more than $\Delta F_{\max} = 0.5$ (periodic cells), or any atom moves by more than $0.35\max(L_x, L_y)$ (finite specimens, Section 7). The guard is checked after the continuous relaxation, in which case the reconnection cascade is skipped, and again after the cascade. When it triggers, the configuration, mesh and cell at the start of the step are restored and written as the final result, and the limit load is reported as lying in $(\alpha_{n-1}, \alpha_n]$.

6.5 Stability of the relaxed states

Under strain control the cell is fixed and a relaxed state is stable when the atomic stiffness $\mathbf{K}^{xx}$ is positive definite on the fluctuations (the two rigid translations excluded). The code monitors this through the lowest eigenvalues of $\mathbf{K}^{xx}$, computed by shift-invert Lanczos at selected steps; $\lambda_{\min}(\mathbf{K}^{xx}) \to 0$ announces an avalanche, and near a saddle-node $\lambda_{\min}^2$ is linear in the distance to the critical load.

Under stress control the free cell components are unknowns as well, and the stability condition is that the Hessian (40) of the enthalpy be positive definite. With $\mathbf{K}^{ww}$ positive definite, this holds if and only if the Schur complement $A_0\,\mathbb{K}_{\mathrm{eff}}$ of (41) is positive definite as well. A relaxed state can therefore lose stability in two ways:

- locally, $\lambda_{\min}(\mathbf{K}^{ww}) \to 0$: an atomic mode softens at fixed cell (nucleation or depinning). The result is an avalanche, a strain burst after which a new equilibrium exists at the same load;
- globally, $\lambda_{\min}(\mathbb{K}_{\mathrm{eff}}) \to 0$ while $\mathbf{K}^{ww}$ stays positive definite: the relaxed cell has no stiffness left along some combination of the free components. This is the limit load of Section 6.4, beyond which no equilibrium exists nearby.

The fixed-cell eigenvalue $\lambda_{\min}(\mathbf{K}^{xx})$ detects only the first. A soft atomic mode that couples to the cell strain also lowers $\mathbb{K}_{\mathrm{eff}}$, since the relaxation term in (41) grows like $1/\lambda_{\min}(\mathbf{K}^{ww})$.

Cell stability matrix. At a relaxed state, the free components satisfy $\mathbf{R}(\bar{\mathbf{F}}) = \bar{\mathbf{P}}(\bar{\mathbf{F}}) - \mathbf{P}^{\mathrm{t}}(\bar{\mathbf{F}}) = \mathbf{0}$ on $\mathcal{S}$, with the atoms relaxed. Its Jacobian is

$$\mathbf{J} = \mathbb{K}_{\mathrm{eff}} - \mathbb{L}, \qquad \mathbb{L} = \frac{\partial \mathbf{P}^{\mathrm{t}}}{\partial \bar{\mathbf{F}}}\Big|_{\mathcal{S}\times\mathcal{S}}, \tag{27}$$

a matrix of size $|\mathcal{S}| \le 3$. Under nominal control $\mathbb{L} = \mathbf{0}$ and $\mathbf{J} = \mathbb{K}_{\text{eff}}$ is symmetric. Under Cauchy control $\mathbb{L}$ is the load stiffness of the follower load, of order $|\boldsymbol{\sigma}^{\text{t}}|$, and $\mathbf{J}$ is not symmetric. In both cases the equilibrium branch folds when a real eigenvalue of $\mathbf{J}$ crosses zero, and we monitor

$$\lambda_{\text{box}} = \min \operatorname{Re} \operatorname{eig} \mathbf{J}. \tag{28}$$

Under Cauchy control $\det \mathbf{J} = 0$ is also the point where $\mathbb{K}_{\text{eff}}^{-1}\mathbb{L}$ has the eigenvalue 1, i.e. where the fixed-point iteration of Algorithm 1 stops contracting (Eq. (22)). $\mathbf{J}$ also gives the slope of the load–strain curve. Write the target as $\mathbf{P}^{\text{t}}(\bar{\mathbf{F}}, \alpha)$, with $\partial \mathbf{P}^{\text{t}}/\partial\alpha = \mathbf{e}_r \otimes \mathbf{e}_c$ for a nominal preset controlling $\bar{P}_{rc}$ and $\partial \mathbf{P}^{\text{t}}/\partial\alpha = \bar{J}\,\boldsymbol{\Sigma}\,\bar{\mathbf{F}}^{-\mathsf{T}}$ for a Cauchy preset with $\boldsymbol{\sigma}^{\text{t}} = \alpha\boldsymbol{\Sigma}$. Differentiating $\mathbf{R} = \mathbf{0}$ along the path gives

$$\frac{\mathrm{d}\bar{\mathbf{F}}}{\mathrm{d}\alpha} = \mathbf{J}^{-1}\frac{\partial \mathbf{P}^{\text{t}}}{\partial \alpha}, \qquad \frac{\mathrm{d}\alpha}{\mathrm{d}\varepsilon} = \Big[\Big(\mathbf{J}^{-1}\frac{\partial \mathbf{P}^{\text{t}}}{\partial \alpha}\Big)_{rc}\Big]^{-1}, \tag{29}$$

with $\varepsilon = \bar{F}_{rc}$ the controlled cell component (Table 1).

Evaluation. Because $\mathbf{x} = \bar{\mathbf{F}}\mathbf{w}$ is linear in $\mathbf{w}$, with $\mathbf{D} = \bar{\mathbf{F}} \otimes \mathbf{I}_n$ we have $\mathbf{K}^{ww} = \mathbf{D}^{\mathsf{T}}\mathbf{K}^{xx}\mathbf{D}$ and $\mathbf{K}^{wF} = \mathbf{D}^{\mathsf{T}}\mathbf{K}^{xF}$. Here $\mathbf{K}^{xF}$ collects the first terms of (38), $\sum_e A_e\, \mathbb{A}_e : (\partial \mathbf{F}_e/\partial \mathbf{x}) \otimes (\partial \mathbf{F}_e/\partial \bar{\mathbf{F}})$. The geometric term of (38) is the atomic force and vanishes at equilibrium. Hence

$$A_0\, \mathbb{K}_{\text{eff}} = \mathbf{K}^{FF} - (\mathbf{K}^{xF})^{\mathsf{T}}(\mathbf{K}^{xx})^{+}\mathbf{K}^{xF}, \tag{30}$$

which needs one sparse factorization of $\mathbf{K}^{xx}$ and $|\mathcal{S}|$ solves. The two translations span the null space of $\mathbf{K}^{xx}$, and the columns of $\mathbf{K}^{xF}$ are orthogonal to it (the forces on a periodic cell sum to zero). The pseudo-inverse in (30) is therefore obtained exactly by pinning one atom. Unlike the preconditioner of Section 8.4, the element tangents are not projected, and the factorization is an $\mathbf{LDL}^{\mathsf{T}}$ decomposition, valid for indefinite matrices. $\mathbb{L}$ is the derivative of $\bar{J}\boldsymbol{\sigma}^{\text{t}}\bar{\mathbf{F}}^{-\mathsf{T}}$ with respect to the free components and is evaluated by central differences. Whenever $\lambda_{\min}(\mathbf{K}^{xx})$ is computed, the code also evaluates λ_{box}, ε and $\mathrm{d}\alpha/\mathrm{d}\varepsilon$. The refinement rule of the monitor also follows λ_{box}, so the approach $\lambda_{\text{box}}^2 \propto \alpha_c - \alpha$ to the limit load is resolved step by step.

Examples. For a perfect 20×20 square cell under nominal shear ($\Delta\alpha = 0.005$, all three components free, normal stresses zero), λ_{box} falls from 2.18 at $\alpha = 0.25$ to 0.97, 0.59 and 0.31 at $\alpha = 0.300$, 0.310 and 0.315. The runaway guard places the limit load in $(0.315, 0.320]$. Extrapolating λ_{box}^2 linearly through the last two states gives $\alpha_c \approx 0.317$. On the elastic segments of a defected 40×40 cell under Cauchy shear, $\mathrm{d}\alpha/\mathrm{d}\varepsilon$ from (29) agrees with the secant slope of the logged curve to better than 1%. λ_{box} stays between 2.1 and 3.5 up to $\sigma_{xy} = 0.118$ (no limit load), while the avalanches are announced by $\lambda_{\min}(\mathbf{K}^{xx})$.

7 Finite specimens: boundary tractions and force grips

7.1 Dead-load tractions

For a finite specimen with boundary $\partial\Omega_0$, let $\Gamma_t \subseteq \partial\Omega_0$ carry the nominal traction $\mathbf{t}(\alpha) = \alpha\,\mathbf{t}_0$, measured per unit reference length and with fixed direction (dead load). The boundary edges of the reference mesh are the edges belonging to exactly one triangle; each carries an outward unit normal $\mathbf{n}$, oriented away from the opposite vertex of its triangle. For linear elements and a traction constant on the edge, the consistent nodal forces are half the edge resultant at each end, so

$$\mathbf{f}_{0,a} = \sum_{\text{edges } (ab)\subset\Gamma_t} \frac{L_{ab}}{2}\,\mathbf{t}_0(\text{edge}), \tag{31}$$

with L_{ab} the reference edge length. Four traction distributions are available:

- Cartesian: $\mathbf{t}_0 = (t_x, t_y)$;
- radial outward or inward: $\mathbf{t}_0 = \pm\mathbf{n}$ on outer contours and $\mp\mathbf{n}$ on hole faces, i.e. always away from (toward) the pivot of the contour;
- circumferential: $\mathbf{t}_0 = \mathbf{e}_z \times \mathbf{r}/|\mathbf{r}|$, with $\mathbf{r}$ the vector from the pivot of the contour (the specimen centre, or the centre of the hole) to the edge midpoint (torsion).

When the loaded region is an edge strip (top, bottom, left, right), only the boundary edges whose normal points in the direction of that face ($\mathbf{n} \cdot \mathbf{d} \geq \frac{1}{2}$) are loaded; the short side edges that a thick strip captures near the corners stay traction-free. Since $\mathbf{n}$ is the reference normal, the radial loading is a dead load, not a follower pressure.

The potential energy is

$$\Pi(\mathbf{x}; \alpha) = E_{\mathrm{el}}(\mathbf{x}) - \alpha \sum_a \mathbf{f}_{0,a} \cdot (\mathbf{x}_a - \mathbf{X}_a), \tag{32}$$

summed over the atoms with free degrees of freedom, so that equilibrium of each free atom reads

$$\frac{\partial E_{\mathrm{el}}}{\partial \mathbf{x}_a} = \alpha\, \mathbf{f}_{0,a}. \tag{33}$$

On atoms with one constrained component (rollers) the constrained component of the gradient is masked after the external force has been added, so that the reaction is carried by the support. As for the periodic cell, $\nabla^2\Pi = \nabla^2 E_{\mathrm{el}}$ and the stability under dead load is that of the elastic Hessian on the free degrees of freedom.

Rigid-body modes and the isostatic anchor. If the specimen has no kinematic boundary condition at all, Π is invariant under the rigid-body motions $\mathbf{x}_a \mapsto \mathbf{Q}\mathbf{x}_a + \mathbf{c}$ only if the load is self-equilibrated,

$$\mathbf{R} = \sum_a \mathbf{f}_{0,a} = \mathbf{0}, \qquad M = \sum_a (\mathbf{X}_a - \mathbf{X}_c) \times \mathbf{f}_{0,a} = 0. \tag{34}$$

Otherwise Π is unbounded below and the problem has no solution; such a load is rejected (with tolerances $|\mathbf{R}| \leq 10^{-10} \sum_a |\mathbf{f}_{0,a}|$ and $|M| \leq 10^{-10} \max(L_x, L_y) \sum_a |\mathbf{f}_{0,a}|$). A self-equilibrated load still leaves the three rigid-body modes as zero-energy directions; they are removed by a minimal isostatic anchor: the atom closest to the centroid is fixed ($u_x = u_y = 0$), and the atom on the same lattice row that lies farthest from it has $u_y = 0$, which removes the rotation. Because the load is balanced, the anchor reactions vanish at equilibrium and the anchor does not perturb the solution.

Supports can also leave some rigid-body modes free. Rollers hold one displacement component, so a translation along x (y) remains free when no atom has u_x (u_y) prescribed. A rotation about (x_0, y_0) remains free when every atom with u_x prescribed lies on the line $y = y_0$ and every atom with u_y prescribed on $x = x_0$, for instance a single row of rollers on one edge, or a single pinned atom. The load must then have no component along the free modes: zero resultant along a free translation, and zero moment about (x_0, y_0) when the rotation is free. Such a load is rejected with the same tolerances.

Conjugate strain. The work-conjugate strain of the dead load is

$$\varepsilon = \frac{1}{A_{\mathrm{s}}} \sum_a \mathbf{f}_{0,a} \cdot (\mathbf{x}_a - \mathbf{X}_a), \tag{35}$$

where $A_{\mathrm{s}} = \sum_e |T_e|$ is the reference solid area. It satisfies $\alpha\, \mathrm{d}\varepsilon = \mathrm{d}W_{\mathrm{ext}}/A_{\mathrm{s}}$, so (α, ε) is a work-conjugate pair. For a rectangle $L_x \times L_y$ loaded by $\mathbf{t}_0 = \mathbf{e}_y$ on the top face and $-\mathbf{e}_y$ on the bottom face, the external work per unit α is $L_x(u_y^{\mathrm{top}} - u_y^{\mathrm{bot}})$ and $\varepsilon = (u_y^{\mathrm{top}} - u_y^{\mathrm{bot}})/L_y$ reduces to the engineering strain. Strain bursts are detected with (26).

7.2 Rigid load-controlled grips

A rigid grip makes all atoms of a region $\mathcal{G}$ move together, $\mathbf{x}_a = \mathbf{X}_a + \mathbf{U}$ for $a \in \mathcal{G}$, with the common displacement $\mathbf{U}$ an additional unknown, and applies the total force $\alpha\mathbf{F}_0$ with $\mathbf{F}_0 = L_{\text{face}}\,(t_x, t_y)$, L_{face} being the reference length of the grip's boundary edges facing outward. The potential and the equilibrium of the grip are

$$\Pi(\mathbf{x}, \mathbf{U}; \alpha) = E_{\text{el}}(\mathbf{x}, \mathbf{U}) - \alpha\,\mathbf{F}_0 \cdot \mathbf{U}, \qquad \frac{\partial \Pi}{\partial \mathbf{U}} = \sum_{a \in \mathcal{G}} \frac{\partial E_{\text{el}}}{\partial \mathbf{x}_a} - \alpha \mathbf{F}_0 = \mathbf{0}, \tag{36}$$

i.e. the resultant of the internal forces on the gripped atoms balances the applied force, while the distribution of the load over the grip is whatever keeps the grip rigid. A component of $\mathbf{U}$ that carries no load can either slide freely or be held at its current value. A grip only transmits the load, so another kinematic support (clamped, prescribed, affine or roller) is required to react it. The conjugate strain is $\varepsilon = \mathbf{F}_0 \cdot \mathbf{U}/A_{\text{s}}$.

Compared with distributed dead-load tractions, the grip imposes a uniform displacement of the loaded face, as in a mechanical test, and is not sensitive to the local compliance of the face near the corners.

7.3 Predictor

Pure force control has no affine predictor. The free atoms (and the grip displacements) are extrapolated from the last two converged states, $\mathbf{x}^{\text{pred}} = \mathbf{x}^{(n)} + \frac{\Delta\alpha_{n+1}}{\Delta\alpha_n}(\mathbf{x}^{(n)} - \mathbf{x}^{(n-1)})$, when step n was smooth, with roller components extrapolated only along their free direction.

8 Newton-type alternatives for the periodic cell

The method of Sections 3–4 uses first derivatives only: the cell components are additional unknowns of an L-BFGS minimization, and the Cauchy target is reached by a fixed-point iteration on the nominal target. This section sets out Newton-type alternatives that use second derivatives of the energy, and compares them with the present method. Variant C, the stiffness-preconditioned minimizer, is implemented and is described in detail in Section 8.4; variant A with the reference tangent was implemented in an earlier stress-controlled example of the code. The others are options.

8.1 Second derivatives of the cell energy

Let $\mathbb{A}_e = \partial^2 W/\partial\mathbf{F}\partial\mathbf{F}$ at $\mathbf{F}_e$, with components $\mathbb{A}_{e,iJkL}$. From $\mathbf{F}_e = \bar{\mathbf{F}}\mathbf{G}_e$ (Eq. (6)) and $\partial\mathbf{F}_e/\partial w_{a,k} = \bar{\mathbf{F}}\,\mathbf{e}_k \otimes \nabla N^e_a$, the blocks of the Hessian of $E_{\text{el}}(\mathbf{w}, \bar{\mathbf{F}})$ are

$$K^{FF}_{ij,kl} = \frac{\partial^2 E_{\text{el}}}{\partial \bar{F}_{ij}\,\partial \bar{F}_{kl}} = \sum_e A_e\,\mathbb{A}_{e,iJkL}\,G_{e,jJ}\,G_{e,lL}, \tag{37}$$

$$K^{Fw}_{ij,ak} = \frac{\partial^2 E_{\text{el}}}{\partial \bar{F}_{ij}\,\partial w_{a,k}} = \sum_{e \ni a} A_e \Big(\mathbb{A}_{e,iJmL}\,\bar{F}_{mk}\,\nabla N^e_{a,L}\,G_{e,jJ} + P_{e,iJ}\,\delta_{jk}\,\nabla N^e_{a,J}\Big), \tag{38}$$

$$\mathbf{K}^{ww}_{ab} = \frac{\partial^2 E_{\text{el}}}{\partial \mathbf{w}_a\,\partial \mathbf{w}_b} = \bar{\mathbf{F}}^{\mathsf{T}}\,\mathbf{K}^{xx}_{ab}\,\bar{\mathbf{F}}, \tag{39}$$

where $\mathbf{K}^{xx}$ is the usual assembled stiffness with respect to positions. The second term of (38) is geometric: it comes from the dependence of $\mathbf{G}_e$ on $\mathbf{w}$. The elastic Hessian with respect to $\mathbf{w}$ and the free components of $\bar{\mathbf{F}}$ is the block matrix

$$\mathbf{K} = \begin{pmatrix} \mathbf{K}^{ww} & \mathbf{K}^{wF} \\ \mathbf{K}^{Fw} & \mathbf{K}^{FF} \end{pmatrix}, \qquad \mathbf{K}^{wF} = (\mathbf{K}^{Fw})^{\mathsf{T}}. \tag{40}$$

Relaxed and Cauchy–Born tangents. Eliminating the atoms at equilibrium gives the relaxed (effective) tangent of the cell, the Schur complement

$$A_0\,\mathbb{K}_{\text{eff}} = \mathbf{K}^{FF} - \mathbf{K}^{Fw}\,(\mathbf{K}^{ww})^{-1}\,\mathbf{K}^{wF}, \qquad \mathrm{d}\bar{\mathbf{P}} = \mathbb{K}_{\text{eff}} : \mathrm{d}\bar{\mathbf{F}} \;\text{ at }\; \partial E_{\text{el}}/\partial \mathbf{w} = \mathbf{0}. \tag{41}$$

$\mathbf{K}^{FF}/A_0$ alone is the affine (unrelaxed) tangent. For a homogeneously deformed lattice all triangles share the same $\mathbf{F}_e$, $\mathbf{P}_e$ and $\mathbb{A}_e$; since a periodic fluctuation $\delta\mathbf{w}$ changes the element gradients by amounts whose area average over the cell vanishes, $\mathbf{K}^{Fw}\delta\mathbf{w} = \mathbf{0}$, the relaxation term vanishes, and $\mathbb{K}_{\text{eff}}$ reduces to the Cauchy–Born tangent $\mathbb{A}(\bar{\mathbf{F}}\mathbf{F}_0)$, the quantity computed by the acoustic-tensor routines of the code ($\mathbf{F}_0$ maps $\mathbb{Z}^2$ to the lattice). In a cell with defects, $\mathbf{K}^{ww}$ is positive definite at a stable equilibrium, so the relaxation term is positive semidefinite and

$$\mathbb{K}_{\text{eff}} \;\preceq\; \mathbf{K}^{FF}/A_0 : \tag{42}$$

the affine tangent is always stiffer than the relaxed one, and the difference grows with the defect density. At the limit load $\mathbb{K}_{\text{eff}}$ becomes singular.

The acoustic tensor $Q_{ik}(\mathbf{n}) = \mathbb{C}_{ijkl}n_j n_l$ built from the spatial Cauchy–Born tangent $\mathbb{C}$ gives the local criterion for loss of strong ellipticity of the homogeneous lattice, $\min_{\mathbf{n}} \det \mathbf{Q}(\mathbf{n}) = 0$. It approximates the onset of homogeneous nucleation in a perfect crystal. The stability of a defected cell under stress control is the global condition that $\mathbf{K}$ of (40) be positive definite. Under a Cauchy target this condition is modified by the load stiffness $-A_0\,\mathbb{L}$, $\mathbb{L} = \partial\mathbf{P}^{\mathrm{t}}/\partial\bar{\mathbf{F}}$.

8.2 Variant A: Newton iteration on the cell only (nested)

The atoms are relaxed at fixed $\bar{\mathbf{F}}$, and the cell is updated by a Newton step on the macroscopic residual $\mathbf{r} = \bar{\mathbf{P}} - \mathbf{P}^{\mathrm{t}}(\bar{\mathbf{F}})$ (components in $\mathcal{S}$):

$$\bar{\mathbf{F}}_{k+1} = \bar{\mathbf{F}}_k - \mathbb{T}^{-1} : \mathbf{r}_k, \tag{43}$$

followed by an affine map of the atoms by $\bar{\mathbf{F}}_{k+1}\bar{\mathbf{F}}_k^{-1}$ and a new relaxation. The exact Jacobian of $\mathbf{r}$ is $\mathbb{J} = \mathbb{K}_{\text{eff}} - \mathbb{L}$. Linearizing about the solution, the error obeys $\mathbf{e}_{k+1} = (\mathbb{I} - \mathbb{T}^{-1}\mathbb{J})\,\mathbf{e}_k$. The choice of $\mathbb{T}$ decides the convergence:

- $\mathbb{T}$ = Cauchy–Born tangent at the reference state ($\mathbf{C} = \mathbf{I}$), used with an under-relaxation factor of 0.1 and a cap of 0.01 on the strain correction in the earlier stress-controlled example of the code. The tangent does not follow the state, and the iteration converges only linearly, under strong damping.

- $\mathbb{T}$ = Cauchy–Born tangent at the current cell, $\mathbb{A}(\bar{\mathbf{F}}_k\mathbf{F}_0)$ with the reduced metric. This is exact for a perfect crystal (quadratic convergence). In a defected cell (42) makes $\mathbb{T}$ too stiff: the iteration converges linearly with ratio $1 - \lambda_{\min}(\mathbb{T}^{-1}\mathbb{J})$, which tends to 1 as the limit load is approached and $\mathbb{K}_{\text{eff}} \to 0$.

- $\mathbb{T} = \mathbb{J}$, with $\mathbb{K}_{\text{eff}}$ from (41). The Schur complement needs three solves with $\mathbf{K}^{ww}$ (one per free component), which one sparse Cholesky factorization provides. Convergence is quadratic and the Cauchy load stiffness is included, so no separate fixed-point loop is needed.

In every case the scheme is nested: each Newton step on the cell requires a complete relaxation of the atoms. A remeshing cascade inside the loop has no single functional to be accepted against, since the cell is updated by a root-finding step and not by minimization.

8.3 Variant B: full Newton on atoms and cell

The atoms and the cell are updated together by

$$\begin{pmatrix} \mathbf{K}^{ww} & \mathbf{K}^{wF} \\ \mathbf{K}^{Fw} & \mathbf{K}^{FF} - A_0\,\mathbb{L} \end{pmatrix} \begin{pmatrix} \delta\mathbf{w} \\ \delta\bar{\mathbf{F}} \end{pmatrix} = - \begin{pmatrix} \partial E_{\mathrm{el}}/\partial \mathbf{w} \\ A_0(\bar{\mathbf{P}} - \mathbf{P}^{\mathrm{t}}) \end{pmatrix}. \tag{44}$$

For a nominal target $\mathbb{L} = 0$, and (44) is the Newton step for the enthalpy H; for a Cauchy target, the load stiffness makes the matrix non-symmetric, and the follower load is treated without an outer loop. Near a stable equilibrium the convergence is quadratic. The difficulty is the regime this model is built to study: plastic events occur exactly where $\mathbf{K}$ loses positive definiteness, and there the Newton step is not a descent direction and may converge to a saddle point or to an unloading branch. A robust version needs globalization, for example a trust region with a truncated conjugate-gradient inner solver, or a modified factorization $\mathbf{K} + \mu\mathbf{I}$ with μ large enough to make it positive definite. With such safeguards the method becomes a damped Newton minimizer of H, and the reconnection cascade can again be accepted on H.

8.4 Variant C: stiffness-preconditioned L-BFGS including the cell

The present minimizer can be given Newton-like steps while keeping descent. This variant is implemented and is used whenever the stiffness preconditioner is selected; this subsection describes it in full.

Change of variables. Let $\mathbf{z} = (\mathbf{w}, \boldsymbol{\zeta})$, $\boldsymbol{\zeta} = \sqrt{A_0}\,(\bar{F}_{ij})_{(ij)\in\mathcal{S}}$, be the unknowns of Section 5, and let $\mathbf{P} = \mathbf{R}\mathbf{R}^{\mathsf{T}}$ be a symmetric positive definite matrix. L-BFGS is run, unmodified, on the variables $\mathbf{t}$ defined by

$$\mathbf{z} = \mathbf{z}_0 + \mathbf{R}^{-\mathsf{T}}\mathbf{t}, \qquad \frac{\partial H}{\partial \mathbf{t}} = \mathbf{R}^{-1}\frac{\partial H}{\partial \mathbf{z}}, \tag{45}$$

starting from $\mathbf{t} = \mathbf{0}$. This is exactly L-BFGS on $\mathbf{z}$ with initial inverse Hessian $\mathbf{P}^{-1}$. If $\mathbf{P}$ is close to the Hessian of H, the Hessian in $\mathbf{t}$ is close to the identity: the first step is a (modified) Newton step, and the number of iterations no longer grows with the condition number of the stiffness, which for plain L-BFGS scales with the square of the cell size in lattice spacings (the long-wavelength modes). The line search is unchanged, so every iterate still lowers H.

The preconditioner. $\mathbf{P}$ is the Gauss–Newton part of the Hessian (40), built from element tangents projected to positive semidefinite form. For each triangle, the eigenvalues of the 4×4 tangent $\mathbb{A}_e$ (in Voigt-like index pairs iJ) are clamped at zero, giving $\mathbb{A}_e^{+}$. With $\mathbf{J}_e = \partial\mathbf{F}_e/\partial(\mathbf{x}, \bar{\mathbf{F}}_{\mathcal{S}})$, the assembled blocks are

$$\begin{aligned} \mathbf{K}^{xx} &= \sum_e A_e \left(\frac{\partial \mathbf{F}_e}{\partial \mathbf{x}}\right)^{\mathsf{T}} \mathbb{A}_e^{+} \frac{\partial \mathbf{F}_e}{\partial \mathbf{x}}, \qquad K^{xF}_{ai,(rc)} = \sum_{e\ni a} A_e \,\frac{\partial \mathbf{F}_e}{\partial x_{a,i}} : \mathbb{A}_e^{+} : \big(\mathbf{e}_r \otimes \mathbf{G}_e^{\mathsf{T}}\mathbf{e}_c\big), \\ K^{FF}_{(rc),(sd)} &= \sum_e A_e \,\big(\mathbf{e}_r \otimes \mathbf{G}_e^{\mathsf{T}}\mathbf{e}_c\big) : \mathbb{A}_e^{+} : \big(\mathbf{e}_s \otimes \mathbf{G}_e^{\mathsf{T}}\mathbf{e}_d\big), \end{aligned} \tag{46}$$

where $\partial\mathbf{F}_e/\partial\bar{F}_{rc} = \mathbf{e}_r \otimes \mathbf{G}_e^{\mathsf{T}}\mathbf{e}_c$ follows from $\mathbf{F}_e = \bar{\mathbf{F}}\mathbf{G}_e$. The geometric term of (38) and the Cauchy load stiffness $-A_0\mathbb{L}$ are left out, so $\mathbf{P}$ is symmetric and positive semidefinite by construction. Since $\mathbf{x} = \bar{\mathbf{F}}\mathbf{w}$, the atomic block in the unknowns $\mathbf{w}$ is obtained with $\mathbf{D} = \bar{\mathbf{F}} \otimes \mathbf{I}_n$ (in the solver layout $[u_1 \dots u_n, v_1 \dots v_n]$), and the scaling $\boldsymbol{\zeta} = \sqrt{A_0}\,\bar{\mathbf{F}}_{\mathcal{S}}$ divides the cell rows by $\sqrt{A_0}$:

$$\mathbf{P} = \begin{pmatrix} \mathbf{M}_{ww} & \mathbf{M}_{w\zeta} \\ \mathbf{M}_{\zeta w} & \mathbf{M}_{\zeta\zeta} \end{pmatrix}, \quad \mathbf{M}_{ww} = \mathbf{D}^{\mathsf{T}}\mathbf{K}^{xx}\mathbf{D} + s\,\mathbf{I}, \quad \mathbf{M}_{w\zeta} = \frac{\mathbf{D}^{\mathsf{T}}\mathbf{K}^{xF}}{\sqrt{A_0}}, \quad \mathbf{M}_{\zeta\zeta} = \frac{\mathbf{K}^{FF}}{A_0}. \tag{47}$$

The shift $s = 10^{-6}\,\overline{\operatorname{diag}\mathbf{M}_{ww}}$ removes the two rigid translations of the periodic cell, which are zero modes of $\mathbf{K}^{xx}$ (it is increased a hundredfold whenever a factorization fails).

Bordered Cholesky factorization. The atomic block is large and sparse, the cell block has at most three rows. The factor is taken in block lower-triangular form,

$$\mathbf{R} = \begin{pmatrix} \mathbf{R}_w & \mathbf{0} \\ \mathbf{B}^\mathsf{T} & \mathbf{L}_S \end{pmatrix}, \qquad \mathbf{R}_w\mathbf{R}_w^\mathsf{T} = \mathbf{M}_{ww}, \quad \mathbf{B} = \mathbf{R}_w^{-1}\mathbf{M}_{w\zeta}, \quad \mathbf{L}_S\mathbf{L}_S^\mathsf{T} = \mathbf{S} = \mathbf{M}_{\zeta\zeta} - \mathbf{B}^\mathsf{T}\mathbf{B}, \tag{48}$$

which satisfies $\mathbf{R}\mathbf{R}^\mathsf{T} = \mathbf{P}$. $\mathbf{R}_w$ is a sparse supernodal Cholesky factor with a fill-reducing ordering, computed once per refresh; $\mathbf{B}$ requires one triangular solve per free cell component; and $\mathbf{S}$, the Schur complement of the atoms in $\mathbf{P}$, is the preconditioner's estimate of the relaxed cell stiffness (the analogue of (41)), a dense matrix of size at most 3×3. It is positive semidefinite because $\mathbf{P}$ is; a small relative shift is added before its dense Cholesky factorization. The two operations needed by (45) are then

$$\mathbf{R}^{-1}\mathbf{g}: \quad \mathbf{y}_w = \mathbf{R}_w^{-1}\mathbf{g}_w, \qquad \mathbf{y}_\zeta = \mathbf{L}_S^{-1}\big(\mathbf{g}_\zeta - \mathbf{B}^\mathsf{T}\mathbf{y}_w\big), \tag{49}$$

$$\mathbf{R}^{-\mathsf{T}}\mathbf{t}: \quad \mathbf{d}_\zeta = \mathbf{L}_S^{-\mathsf{T}}\mathbf{t}_\zeta, \qquad \mathbf{d}_w = \mathbf{R}_w^{-\mathsf{T}}\big(\mathbf{t}_w - \mathbf{B}\,\mathbf{d}_\zeta\big), \tag{50}$$

that is, two sparse triangular solves per evaluation of H, plus $O(nm)$ dense operations for the border. The cell is therefore preconditioned through its coupling to all atoms: a change of the cell is accompanied, in the preconditioned metric, by the non-affine relaxation of the atoms that the stiffness predicts.

Refresh, stalls and termination. $\mathbf{P}$ is assembled and factorized at the start of every stress-controlled relaxation, i.e. once per load step under nominal control and once per pass of Algorithm 1 under Cauchy control (or less often, if a reuse interval is set). The symbolic analysis is kept while the mesh is unchanged. During an avalanche the configuration moves far from the point where $\mathbf{P}$ was built, and L-BFGS can stall before reaching the tolerance; $\mathbf{P}$ is then rebuilt at the current point and the minimization continues (at most three times). The relaxation stops when

$$\max\big|\partial H/\partial \mathbf{z}\big| < \varepsilon_g, \tag{51}$$

measured on the unpreconditioned gradient, whose atomic part is the force and whose cell part is $\sqrt{A_0}(\bar{P}_{ij} - P^\mathrm{t}_{ij})$ by (24); with $\varepsilon_g = 10^{-6}$ the macroscopic stress is matched to $\varepsilon_g/\sqrt{A_0}$. In the preconditioned variables the step-length cap of Section 5 is not used, because the first step is already scaled by the stiffness. The limit-load guard of Section 6.4 is unchanged. If a factorization fails, the relaxation falls back to plain L-BFGS.

Performance and path dependence. Table 2 compares the two solvers on a defected periodic cell of 100×100 atoms (square lattice, 2×10^4 triangles), relaxed at $\bar{F}_{12} = 0.14$ with noise and then loaded in Cauchy shear in steps of $\Delta\sigma = 0.005$, with the reconnection cascade enabled. Each preconditioned relaxation takes between 0 and about 40 iterations and at most ~ 0.1 s including assembly and factorization, so that the remaining cost of an elastic load step is dominated by the reconnection trial and the output, not by the solver. Stresses are matched equally well by both solvers ($|\bar{\sigma}_{12} - \sigma^\mathrm{t}_{12}| \lesssim 10^{-8}$, normal components $\lesssim 10^{-9}$).

The two solvers produce the same states, to the solver tolerance, until the first avalanche. During an avalanche they can settle into different metastable states; in the run of seed 42 the preconditioned solver reached a lower energy in the fifth reconnection pass of the first stress step, and the two dislocation structures differ from there on, with the limit load reached at $\sigma = 0.047$ by the plain solver and not before 0.057 by the preconditioned one. This sensitivity is a property of the avalanche dynamics, not of either solver: any change of the minimizer, of its tolerance or of rounding selects among nearby metastable states. Quantities such as the limit load and the avalanche statistics should therefore be reported as ensemble averages over noise realizations.

	seed 42		seed 7	
	plain	preconditioned	plain	preconditioned
total wall time (s)	119	35	134	51
load steps completed	8 (limit load)	10	16	16
elastic step (s)	1.9–3.8	0.5–0.8	1.8–2.8	0.5–1.0
step with avalanche (s)	9–53	4–7	5–29	2–12

Table 2: Plain and stiffness-preconditioned L-BFGS for Cauchy shear of a defected 100×100 periodic cell (same initial state for both solvers within each seed; six threads per run).

8.5 Variant D: accelerating the Cauchy loop

Independently of the inner solver, the Cauchy iteration of Algorithm 1 can be written as a root-finding problem for the cell, $\mathbf{r}(\bar{\mathbf{F}}) = \bar{\mathbf{P}}(\bar{\mathbf{F}}) - \mathbf{P}^{\mathrm{t}}(\bar{\mathbf{F}}) = 0$, with $\bar{\mathbf{P}}(\bar{\mathbf{F}})$ the relaxed stress. A secant (Broyden) update of $\mathbb{J}$ built from the successive passes, or Anderson acceleration of the fixed-point map, turns the linear contraction (22) into superlinear convergence without computing a tangent. Since every pass after the first is warm-started and short, the gain is limited to the passes themselves.

8.6 Comparison

Table 3 compares the variants with the present method (first row).

Method	Jacobian used	Convergence	Robust at instabilities	Remesh acceptance
L-BFGS on $(\mathbf{w}, \bar{\mathbf{F}})$ + fixed point (present)	none (gradient only)	superlinear (inner), linear with ratio $\lvert\boldsymbol{\sigma}^{\mathrm{t}}\rvert/\mu_{\mathrm{eff}}$ (outer)	yes: descent with line search	on H
A: cell Newton, reference CB tangent	$\mathbb{A}(\mathbf{F}_0)$	linear, heavy damping	partly (damping)	no functional
A: cell Newton, current CB tangent	$\mathbb{A}(\bar{\mathbf{F}}\mathbf{F}_0)$	quadratic (perfect crystal); linear, $\to 1$ at limit load (defects)	partly	no functional
A: cell Newton, Schur tangent	$\mathbb{K}_{\mathrm{eff}} - \mathbb{L}$	quadratic (outer); nested relaxations	no near singular $\mathbb{K}_{\mathrm{eff}}$	no functional
B: full Newton	$\mathbf{K}$, $-A_0\mathbb{L}$	quadratic near stable equilibria	only with trust region / shift	on H if globalized
C: preconditioned L-BFGS (implemented)	$\mathbf{K}_+ + s\mathbf{I}$ as preconditioner	near-Newton per iteration, fewer iterations	yes: descent with line search	on H
D: Broyden / Anderson outer	secant estimate of $\mathbb{J}$	superlinear (outer)	as inner solver	unchanged

Table 3: Newton-type alternatives for stress control of the periodic cell. CB: Cauchy–Born. Variants C and D can be combined with each other and with the present formulation.

Cost. A sparse Cholesky factorization of the $2N \times 2N$ atomic stiffness of a two-dimensional mesh costs $O(N^{3/2})$ with nested-dissection ordering, against $O(N)$ for an L-BFGS iteration. The bordering by at most three cell components adds three solves with the factor. For variant C the balance has been measured (Table 2): on a 100×100 defected cell it is a factor of 2.6–3.4 in total run time and 3–5 on elastic steps. For the other variants it has not been measured.

Assessment. The Newton variants pay off in the elastic stretches between plastic events, where the energy is locally convex and quadratic convergence is available. They give no advantage at the plastic events themselves, where the Hessian is indefinite and only a descent method selects the physically relevant branch. Variant A with a Cauchy–Born tangent, current or not, is limited by (42) in exactly the defected states of interest. Variants C and D keep the present formulation, including the energy acceptance of the reconnection cascade and the detection of the limit load, and add second-order information only where it is safe to do so.

9 Summary

	unknowns	functional	equilibrium
PBC, nominal stress	$\mathbf{w}$, $\bar{F}_{ij\in\mathcal{S}}$	$E_{\mathrm{el}} - A_0\mathbf{P}^{\mathrm{t}} : (\bar{\mathbf{F}} - \mathbf{I})$	$\bar{P}_{ij} = P^{\mathrm{t}}_{ij}$
PBC, Cauchy stress	$\mathbf{w}$, $\bar{F}_{ij\in\mathcal{S}}$	frozen-target H, fixed point	$\bar{P}_{ij} = [\bar{J}\boldsymbol{\sigma}^{\mathrm{t}}\bar{\mathbf{F}}^{-\mathsf{T}}]_{ij} \Rightarrow \bar{\boldsymbol{\sigma}} = \boldsymbol{\sigma}^{\mathrm{t}}$
boundary traction	free $\mathbf{x}_a$	$E_{\mathrm{el}} - \alpha\sum_a \mathbf{f}_{0,a}\cdot(\mathbf{x}_a - \mathbf{X}_a)$	$\partial E_{\mathrm{el}}/\partial\mathbf{x}_a = \alpha\mathbf{f}_{0,a}$
rigid grip	free $\mathbf{x}_a$, $\mathbf{U}$	$E_{\mathrm{el}} - \alpha\mathbf{F}_0\cdot\mathbf{U}$	$\sum_{\mathcal{G}}\partial E_{\mathrm{el}}/\partial\mathbf{x}_a = \alpha\mathbf{F}_0$

Nominal stresses and tractions defined per unit reference length are dead loads: they derive from a work potential linear in the kinematic unknowns, and equilibria are local minima of an enthalpy whose Hessian is the elastic one. A prescribed Cauchy stress is a follower load without a potential; it is enforced exactly, in all four components, by iterating the dead-load problem on the nominal target $\bar{J}\boldsymbol{\sigma}^{\mathrm{t}}\bar{\mathbf{F}}^{-\mathsf{T}}$, a contraction with ratio of order $|\boldsymbol{\sigma}^{\mathrm{t}}|/\mu_{\mathrm{eff}}$ that fails only at the limit load. In all cases avalanches are identified as bursts of the work-conjugate strain at fixed load, and the run stops at the limit load while keeping the last equilibrium state.